Status	Finished
Started	Friday, 16 October 2026, 11:08 AM
Completed	Friday, 16 October 2026, 11:12 AM
Duration	4 mins 1 sec
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

a. Bubble Sort does not guarantee sorted order
b. Bubble Sort is less efficient for large lists
c. Bubble Sort cannot handle duplicate elements
d. Bubble Sort is difficult to implement

The correct answer is: Bubble Sort is less efficient for large lists

Question **2**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Sorting
b. Searching
c. Complexity
d. Merging

The correct answer is:
Complexity

Question **3**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

- a. By comparing elements sequentially
- b. By using the bubble-up method
- c. By breaking the input into smaller parts and merging them
- d. By sorting data in a single pass

The correct answer is: By breaking the input into smaller parts and merging them

Question **4**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

- a. The parts are merged without sorting
- b. Each part is sorted independently
- c. Each part is searched for a specific element
- d. The parts are ignored

The correct answer is: Each part is sorted independently

Question **5**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

- a. All of the above
- b. Selection
- c. Duplicates
- d. Searching

The correct answer is: All of the above

Question **6**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

- a. Bubbling up the smallest element
- b. Dividing the list into halves
- c. Bubbling up the largest element to its correct position
- d. Sorting the entire list in one pass

The correct answer is: Bubbling up the largest element to its correct position

Question **7**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

- a. Bubble Sort
- b. Linear Search
- c. Binary Search
- d. Merge Sort

The correct answer is: Merge Sort

Question **8**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

- a. Inserting data into a list
- b. Deleting data from a list
- c. Arranging data in a particular format
- d. Searching for data in a list

The correct answer is: Arranging data in a particular format

Question **9**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

- a. Merge Sort
- b. Linear Search
- c. Quick Sort
- d. Bubble Sort

The correct answer is: Merge Sort

Question **10**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

- a. Merge Sort
- b. Insertion Sort
- c. Quick Sort
- d. Bubble Sort

The correct answer is: Bubble Sort

Question **11**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

- a. Dividing the input into smaller subproblems
- b. Sorting the entire list sequentially
- c. Comparing each element with the others
- d. Combining sorted sublists

The correct answer is: Dividing the input into smaller subproblems

Question **12**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

a. Sorting is rarely used in programming
b. Sorting decreases the efficiency of selection operations
c. Sorting helps in finding duplicates quickly
d. Sorting makes it harder to search for items

The correct answer is: Sorting helps in finding duplicates quickly

Question **13**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

a. Quick Sort
b. Merge Sort
c. Heap Sort
d. Bubble Sort

The correct answer is: Bubble Sort

Question **14**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

a. Bubbling up the largest element to its correct position with each pass
b. Being the most efficient sorting algorithm
c. Sorting data in a non-sequential manner
d. Using the divide-and-conquer approach

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **15**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It avoids the need for recursion
b. It allows for efficient parallel processing and sorting of data
c. It simplifies the sorting process by using only one pass
d. It only works on small datasets

The correct answer is: It allows for efficient parallel processing and sorting of data